$$z = \pm\left(\frac{d^2 - \Delta_n^2}{\Delta_n^2}\right)^{1/2} x \qquad \textbf{(14.9)}$$

where $\Delta_n = \left(n + \frac{1}{2}\right)\lambda$; $n = 0, 1, 2,\ldots$Clearly the points at which minima will occur (on the line $z = D$) would be given by

$$x_n = \pm\left(\frac{\Delta_n^2}{d^2 - \Delta_n^2}\right)^{1/2} D$$

$$= \pm\frac{\Delta_n}{d}\left(1 - \frac{\Delta_n^2}{d^2}\right)^{-1/2} D$$

$$\approx \pm\left(n + \frac{1}{2}\right)\frac{\lambda D}{d} \qquad \textbf{(14.10)}$$

where we have assumed $\Delta_n << d$. Thus, the points corresponding to minima will be (approximately) equally spaced with a spacing of $\lambda D/d$.

Example 14.4 Till now we have assumed the needles at S_1 and S_2 (see Fig. 14.1) to vibrate in phase. Assume now that the needles vibrate with a phase difference of π and obtain the nodal lines. Generalize the result for an arbitrary phase difference between the vibrations of the two needles.

Solution: The two needles S_1 and S_2 vibrate out of phase. Thus if, at any instant, the needle at S_1 produces a crest at a distance R from it then the needle at S_2 would produce a trough at a distance R from S_2. Therefore, at all points on the perpendicular bisector OZ (see Fig. 14.4) the two vibrations will always be out of phase and we will have minimum. On the other hand, at the point B which satisfies the equation

$$S_2B - S_1B = \lambda/2$$

the two vibrations will be in phase and we will have maximum. Thus, because of the initial phase difference of π, the conditions for maxima and minima are reversed, i.e., when

$$S_2P \sim S_1P = \left(n + \frac{1}{2}\right)\lambda \text{ (maxima)}$$

the interference will be constructive and we will have maxima, and when

$$S_2P \sim S_1P = n\lambda \text{ (minima)}$$

the interference will be destructive and we will have minima. Notice that one again obtains a stationary interference pattern with nodal lines as hyperbolae.

The above analysis can easily be generalized for arbitrary phase difference between the two needles. Assume, for example, that there is a phase difference of $\pi/3$, i.e., if there is a crest at a distance R from S_1 then there is a crest at a distance $R - \lambda/6$ from S_2. Consequently, the condition

$$S_1P - S_2P = n\lambda + \frac{\lambda}{6};\ n = 0, \pm 1, \pm 2,\ldots$$

will correspond to maxima.

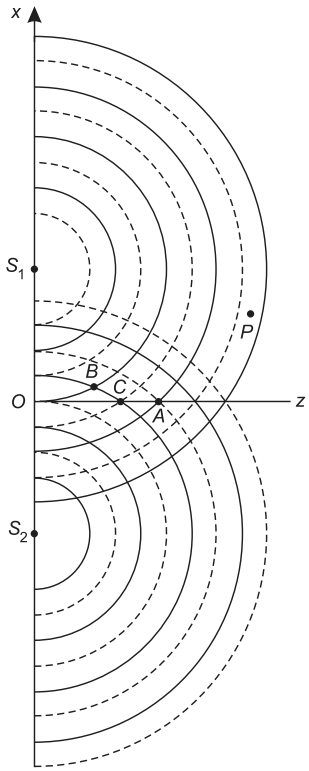


Fig. 14.4 Waves emanating from two point sources S_1 and S_2 vibrating out of phase.

14.3 COHERENCE LO 2

From the above examples, we find that whenever the two needles vibrate with a constant phase difference, a stationary interference pattern is produced. The positions of the maxima and minima will, however, depend on the phase difference in the vibration of the two needles. Two sources which vibrate with a fixed phase difference between them are said to be coherent.

We next assume that the two needles are sometimes vibrating in phase, sometimes vibrating out of phase, sometimes vibrating with a phase difference of $\pi/3$, etc., then the interference pattern will keep on changing. If the phase difference changes with such great rapidity that a stationary interference cannot be observed then the sources are said to be *incoherent*.

Let the displacement produced by the sources at S_1 and S_2 be given by

$$\left.\begin{aligned} y_1 &= a\cos\omega t \\ y_2 &= a\cos(\omega t + \phi) \end{aligned}\right\} \qquad \textbf{(14.11)}$$

then the resultant displacement would be

$$y = y_1 + y_2 = 2a\cos\phi/2\cos(\omega t + \phi/2) \qquad \textbf{(14.12)}$$